

6.4.6 System Qualification Testing Process

The System Qualification Testing Process contributes to demonstrating that the integrated system satisfies its defined system requirements and is ready for delivery and operation.

6.4.6.1 Purpose

The purpose of the System Qualification Testing Process is to ensure that the implementation of each system requirement is tested for compliance and that the system, as integrated, is verified to be ready for delivery in its intended operational environment.

NOTE — This process focuses on qualification of the integrated system as a whole and provides objective evidence that system requirements have been met.

6.4.6.2 Outcomes

As a result of successful implementation of the System Qualification Testing Process:

- a) Criteria are developed for evaluating compliance with all applicable system requirements.
 - b) The integrated system is tested using the defined qualification criteria.
 - c) Qualification test results are recorded and maintained as verification evidence.
 - d) The readiness of the system for delivery is assured based on qualification results and supporting evidence.
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6.4.6.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.4.6.3.1 Qualification Testing

This activity ensures that the integrated system satisfies its defined requirements

and that qualification evidence is collected and documented.

- **6.4.6.3.1.1 Document Qualification Requirements**

Document the system requirements that must be verified during qualification testing. Ensure traceability between requirements, qualification criteria, and test cases.

- **6.4.6.3.1.2 Develop Qualification Criteria**

Define qualification criteria for evaluating compliance with each requirement.

Evaluation criteria should address at least the following:

- a) Test coverage of all system requirements
- b) Conformance to expected results and performance specifications
- c) Feasibility of system operation and maintenance under intended

conditions

Qualification criteria should also address any regulatory, safety, security, or interoperability requirements applicable to the system.

- **6.4.6.3.1.3 Plan and Prepare Qualification Tests**

Develop qualification test plans, procedures, and required resources. Establish and configure the test environment to replicate operational conditions as closely as practicable.

- **6.4.6.3.1.4 Execute Qualification Tests**

Conduct tests of the integrated system according to the qualification plan. Monitor and control test execution to ensure consistency and reliability.

- **6.4.6.3.1.5 Record and Evaluate Results**

Record the results of all qualification tests, including observed behavior, measured values, anomalies, and deviations from expected results. Evaluate the system against the qualification criteria to determine compliance.

- **6.4.6.3.1.6 Perform Audits (if applicable)**

Support and document qualification-related audits in accordance with the Software Audit Process (7.2.7). If audits are conducted, record their results and ensure any non-conformities are addressed.

NOTE — This subclause does not apply to software configuration items for which audits have already been conducted.

- **6.4.6.3.1.7 Confirm Readiness for Delivery**

Based on the results of qualification testing and any related audits, update and prepare the integrated system for Software Installation (6.4.7) and Software Acceptance Support (7.2.8), if applicable. Provide qualification evidence to relevant stakeholders for approval.

Cross-References

This process is closely related to:

- **Software Verification Process (7.2.4)** — for verifying software components against their requirements, which may be incorporated into overall

system qualification.

- **Software Audit Process (7.2.7)** — for auditing qualification evidence and configuration items.
- **Software Installation Process (6.4.7)** — for subsequent installation of the qualified system into the operational environment.